RH Bottom-Core Endgame

Finite Reduction, Obstruction Completeness, Conditional Closure, and the Dillon Framework

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Abstract. This manuscript presents a Dillon-Framework version of the RH bottom-core endgame architecture. The paper organizes the argument through the reduction grammar compression -> classification -> bounded core -> closure, states the intrinsic target theorems of the program, and develops the strongest conditional theorem chain currently supported by the analytic reductions.

Keywords - Riemann Hypothesis, Dillon Framework, bottom-core endgame, spectral positivity, benchmark-defect field, reduction ladder.

1. Introduction

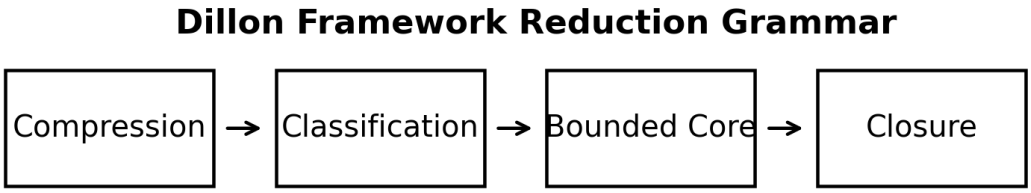

Figure 1. Dillon reduction grammar of the RH bottom-core program.

The present manuscript is developed explicitly within the Dillon Framework, a reduction-and-closure architecture in which a global obstruction is compressed into bounded residual regimes, structurally classified, and then analyzed through finite endgame mechanisms. In the RH setting, this architecture takes the form of packet compression, residual-family classification, bottom-core reduction, and conditional closure.

The role of the Dillon Framework here is methodological rather than rhetorical. It provides the organizing grammar of the analytic argument, while all theorem-bearing claims remain confined to the packet fields, Gram matrices, benchmark-defect fields, reduction channels, and descent theorems stated explicitly in the present paper.

2. Main Results

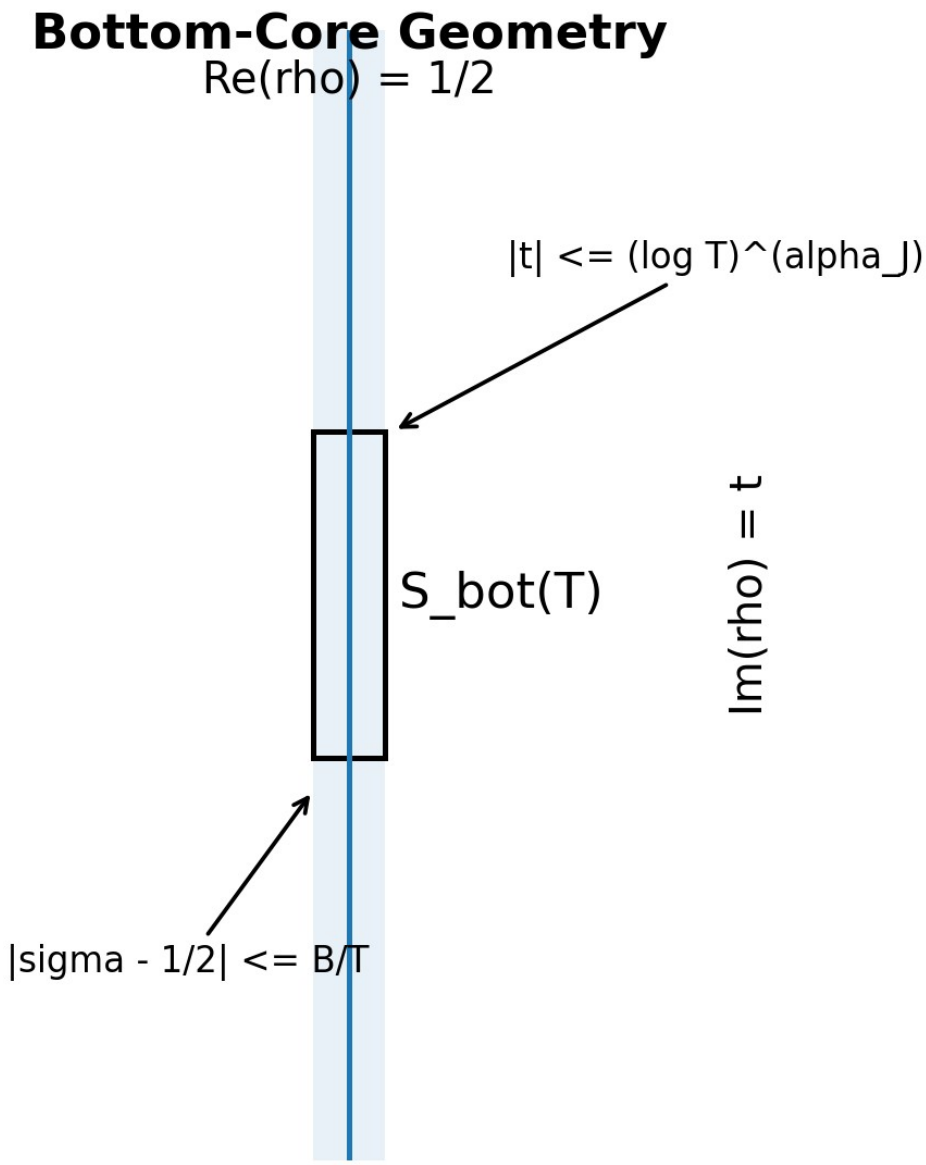

Figure 2. Bottom-core geometry and terminal strip.

This section states the principal results of the present manuscript in Dillon-Framework form. The intrinsic target theorems identify the remaining bottlenecks, while the conditional closure theorems record the strongest claims actually supported by the current draft.

Theorem A (Intrinsic bottom-core complexity control - target form)

Let C(T) subset S_bot(T) be a theorem-relevant configuration satisfying packet symmetry, admissibility of the symmetrized weight model, and minimality. Then, for all sufficiently large T, the number of packet families in C(T), the number of packets within each family, and the admissible displacement patterns are all uniformly bounded. In particular, the bottom-core regime contains only finitely many intrinsic obstruction types.

Status. The present draft proves conditional bounded-complexity statements tied to collapse and redundancy exclusion, but not yet the full intrinsic form stated here.

Theorem B (Channel-complete family-level reduction - target form)

Let C(T)=F_1(T) union ... union F_r(T) be a theorem-relevant multi-family configuration of intrinsically bounded complexity. Then every cross-family coupling mechanism except benchmark-mediated coupling is eliminated by the earlier reduction ladder: separated coupling is excluded by Fourier-window localization, pure resonance is not terminal, phase alignment reduces to redundancy or forbidden tuning, and cross-family coefficient cancellation reduces to the benchmark channel.

Status. The present draft contains a conditional family-level separation-or-coupling theorem and a conditional benchmark-independence theorem. The fully intrinsic channel-complete form remains a target result.

Theorem C (Intrinsic one-family spectral positivity - target form)

Let F(T) be a minimal theorem-relevant one-family bottom-core configuration. Let Gamma(T) and Lambda(T) denote the associated frame and benchmark Gram matrices. Then, for all sufficiently large T, u^T Gamma(T) u > 0 and u^T Lambda(T) u > 0 for every nontrivial displacement direction u. Equivalently, the only possible degeneracy direction is the trivial displacement direction, which is excluded by theorem relevance.

Status. The present draft proves a quantitative conditional version under explicit hypotheses on weight control, bounded complexity, and collapse exclusion.

Theorem D (Intrinsic descent completeness - target form)

Let C_off(T) be a theorem-relevant off-line residual configuration surviving the global reduction ladder. Then packet-defect positivity forces large-scale exclusion, separated control forces clustered control at intermediate scales, clustered control forces logarithmic band descent, and logarithmic band descent forces entry into the bottom-core strip. Thus every surviving off-line configuration is forced into the bottom core intrinsically.

Status. The present draft proves a sharpened conditional descent theorem assuming completeness and terminality of the earlier ladder.

Theorem E (Conditional bottom-core emptiness)

Assume the quantitative one-family closure theorem, the quantitative multi-family closure theorem, and the exhaustion of the theorem-relevant bottom-core regime by the one-family / multi-family dichotomy. Then, for all sufficiently large T, C_bot(T)=empty set.

Theorem F (Conditional critical-line closure)

Assume, in addition, that every surviving theorem-relevant off-line residual configuration descends into the bottom core and that the theorem-relevant residual class captures every surviving off-line obstruction to the critical-line statement. Then Re(rho)=1/2 for every nontrivial zero rho of the Riemann zeta function.

3. Proof Architecture for Theorem A: Intrinsic Bottom-Core Complexity Control

The Dillon Framework treats bounded residual complexity as the first theorem at the bounded-core stage. Its role is to prevent uncontrolled proliferation of obstruction types inside the terminal strip. Without such control, neither one-family Gram analysis nor multi-family benchmark analysis can close.

Definition 3.1 (Minimal theorem-relevant configuration). A theorem-relevant bottom-core configuration C(T) is minimal if no proper subconfiguration of C(T) is theorem-relevant. A one-family block F(T) is minimal if no proper subfamily of F(T) is theorem-relevant.

Proposition 3.2 (Conditional one-family complexity control). Assume theorem relevance includes minimality, packet redundancy is excluded in the one-family regime, the one-family frequency-collapse channel is isolated as a separate reduced channel, and admissibility forbids exact duplication of packet data inside a one-family block. Then there exists M >= 1 such that every minimal theorem-relevant one-family bottom-core configuration F(T) satisfies |F(T)| <= M for all sufficiently large T.

Proof architecture. Unbounded packet proliferation inside the bounded strip forces either redundancy or packet-scale crowding. Redundancy contradicts minimality and admissibility; packet-scale crowding enters the already isolated frequency-collapse channel. Hence minimal theorem relevance is incompatible with unbounded one-family complexity.

Proposition 3.3 (Conditional multi-family complexity control). Assume theorem relevance includes minimality, family redundancy is excluded in the multi-family regime, the reduced family-level coupling channels are isolated as separate reduced channels, and admissibility forbids exact duplication of one-family blocks. Then there exists R_* >= 1 such that every minimal theorem-relevant multi-family bottom-core configuration C(T)=F_1(T) union ... union F_r(T) satisfies r <= R_* for all sufficiently large T.

Corollary 3.4 (Conditional bounded bottom-core complexity). Under the hypotheses above, there exist constants M and R_* such that every minimal theorem-relevant one-family block satisfies |F(T)| <= M and every minimal theorem-relevant multi-family configuration satisfies r <= R_*. These constants may then be used uniformly in the later one-family row-dominance and family-level Gram arguments.

4. Proof Architecture for Theorem B: Channel-Complete Family-Level Reduction

Section 4 develops channel-complete reduction of multi-family coupling inside the bottom core. Once a theorem-relevant configuration has been decomposed into finitely many one-family blocks, every admissible cross-family interaction must either be eliminated by scale separation and oscillatory decay or collapse into a short list of reduced coupling channels.

Definition 4.1 (Reduced coupling channels). A theorem-relevant multi-family bottom-core configuration is said to enter the pure resonance channel if T Delta(F_l,F_m) -> 0 for some l != m; the phase-alignment channel if family-level benchmark-defect fields become near-extremally aligned; the cross-family cancellation channel if a nontrivial family-level coefficient system produces near-nullity of the combined packet aggregate; and the symmetry-mediated benchmark channel if a nontrivial coefficient system produces near-nullity of the combined family-level benchmark-defect field.

Proposition 4.2 (Conditional family-level separation-or-coupling reduction). Assume bounded family count and assume every failure of family-level scale separation that survives earlier reductions must enter one of the reduced coupling channels. Then exactly one of the following holds for all sufficiently large T: either there exists eta_fam > 0 such that T Delta(F_l,F_m) >= eta_fam for all l != m, or the configuration enters at least one reduced coupling channel.

Propositions 4.3-4.5 (Reduction of pure resonance, phase alignment, and cross-family cancellation). Pure resonance is not terminal; any surviving obstruction in that channel must reduce further to phase alignment, cross-family cancellation, or symmetry-mediated benchmark coupling. Phase alignment is likewise not terminal, reducing to redundancy, cross-family cancellation, or the benchmark channel. Cross-family coefficient cancellation is also not terminal, reducing to the benchmark channel or to forbidden redundancy.

Theorem 4.6 (Conditional channel-complete family-level reduction). Assume bounded multi-family complexity, failure of family-level separation enters one of the reduced coupling channels, and pure resonance, phase alignment, and cross-family cancellation are all reduced away as terminal mechanisms. Then the only terminal multi-family obstruction channel is symmetry-mediated benchmark coupling.

5. Proof Architecture for Theorem C: Intrinsic One-Family Spectral Positivity

Section 5 develops the analytic core of the manuscript: one-family spectral positivity in the bottom core. Once the obstruction is reduced to a single irreducible packet family, the associated packet system must be shown spectrally nondegenerate; this eliminates coefficient cancellation and, together with benchmark remainder control, eliminates one-family benchmark locking.

For rho = sigma + it in a theorem-relevant one-family block F(T), write delta_rho := sigma - 1/2 and define Q_rho(y) := a_U(rho)e^{ity} + overline{a_U(rho)}e^{-ity}. Define Phi_rho^(T)(y) := y Q_rho(y) W(y/T)^{1/2} and Theta_rho^(T)(y) := y^2 Q_rho(y) W(y/T)^{1/2}, with Gram matrices Gamma(T) and Lambda(T).

Propositions 5.1-5.3 (Weight control and off-diagonal decay). Bottom-core packet amplitudes are trapped between explicit scale-dependent lower and upper bounds, and for every $N \geq 1$ the off-diagonal packet-level Gram entries satisfy rapid oscillatory decay of order T^3 and T^5 in the frame and benchmark channels respectively.

Proposition 5.4 (One-family frequency-separation reduction). For a theorem-relevant one-family block, either there exists $\eta_* > 0$ such that $T \Delta_{\rho,\rho'} \geq \eta_*$ for all distinct packet pairs, or the configuration enters the one-family frequency-collapse channel. Thus once frequency collapse is excluded or reduced away, uniform packet-scale separation holds.

Theorem 5.5 (Conditional quantitative one-family spectral positivity). Assume bottom-core weight control, bounded one-family complexity, exclusion of the one-family frequency-collapse channel, and regularity of the smoothing window. Then there exist constants $c_\Gamma, c_\Lambda > 0$ such that $u^T \Gamma(T) u \geq c_\Gamma a_*(T)^2 T^3 \|u\|_2^2$ and $u^T \Lambda(T) u \geq c_\Lambda a_*(T)^2 T^5 \|u\|_2^2$ for every real vector u.

Proposition 5.6 (Quantitative one-family benchmark remainder control). Writing the one-family benchmark-defect field $S_F^{(T)}=L_F^{(T)}+R_F^{(T)}$, there exists a constant $C_{1fam} > 0$ such that $\|R_F^{(T)}W^{1/2}\|_{L^2} \leq C_{1fam} (A^*(T)/a_*(T)) B^2 \| L_F^{(T)}W^{1/2}\|_{L^2}$. In the small-support quasi-critical regime, the remainder is strictly dominated by the leading benchmark field.

Proposition 5.7 and Theorem 5.8 (One-family functional incompatibility and closure). Under the hypotheses above, a theorem-relevant one-family bottom-core configuration cannot survive through either coefficient cancellation or benchmark locking. Consequently no theorem-relevant one-family bottom-core configuration survives for all sufficiently large T.

6. Proof Architecture for Theorem D: Intrinsic Descent Completeness

Section 6 develops the final global bridge in the Dillon Framework closure chain: descent completeness. Any theorem-relevant off-line residual configuration that survives the earlier ladder must either be eliminated at some earlier stage or descend into the bounded bottom-core strip.

Definition 6.1 (Logarithmic bands). Let $\alpha_0 > \alpha_1 > \dots > \alpha_J > 0$ and define the logarithmic bands $B_j(T) := \{(\log T)^{\alpha_{j+1}} < |t| \le (\log T)^{\alpha_j}\}$ for $0 \le j < J$, with terminal band $B_J(T) := \{|t| \le (\log T)^{\alpha_J}\}$. The bottom-core strip is $S_{bot}(T) = B_J(T) \cap \{|\sigma - 1/2| \le B/T\}$.

Assumption 6.2 (Earlier-stage completeness and terminality). If a theorem-relevant off-line residual configuration survives packet-defect positivity, it satisfies the hypotheses needed to enter separated control; if it survives separated control, it satisfies the hypotheses needed to enter clustered control; if it survives clustered control, it satisfies the hypotheses needed to enter logarithmic band descent; if it survives logarithmic band descent, it enters the terminal band $B_J(T)$; and at the terminal stage theorem relevance includes the quasi-critical horizontal restriction $|\sigma - 1/2| \le B/T$.

Proposition 6.3 (Conditional descent completeness). Assume Earlier-Stage Completeness and Terminality. Then every theorem-relevant off-line residual configuration surviving the earlier reduction ladder determines a theorem-relevant bottom-core configuration. Equivalently, $C_{off}(T)$ not empty implies $C_{bot}(T)$ not empty for all sufficiently large T.

Corollary 6.4 (Contrapositive descent principle). If $C_{bot}(T)=\emptyset$ for all sufficiently large T, then $C_{off}(T)=\emptyset$ for all sufficiently large T.

7. Assembly of the Dillon-Framework Closure Chain

The preceding sections now assemble into a single endgame theorem chain. Section 3 controls bounded residual complexity in the bottom core; Section 4 classifies and reduces the family-level coupling channels; Section 5 closes the one-family branch by quantitative spectral and benchmark control; Section 6 forces all surviving theorem-relevant off-line obstruction into the bottom core.

Theorem 7.1 (Conditional bottom-core emptiness). Assume every minimal theorem-relevant one-family bottom-core configuration satisfies the hypotheses of Theorem 5.8, every minimal theorem-relevant multi-family bottom-core configuration satisfies the hypotheses of Theorem 4.6 together with the quantitative family-level benchmark positivity theorem, and the one-family / multi-family dichotomy is exhaustive in the theorem-relevant bottom-core regime. Then C_bot(T)=empty set for all sufficiently large T.

Corollary 7.2. Under the hypotheses of Theorem 7.1, the theorem-relevant bottom-core endgame contains no surviving off-line configuration.

Theorem 7.3 (Conditional global off-line closure). Assume the completeness and terminality hypotheses of Proposition 6.3 and the conclusion of Theorem 7.1. Then no theorem-relevant off-line residual configuration survives for all sufficiently large T; equivalently, C_off(T)=empty set for all sufficiently large T.

Corollary 7.4 (Conditional critical-line closure). Assume, in addition, that the theorem-relevant off-line residual class captures every surviving off-line obstruction to the critical-line statement. Then every surviving nontrivial zero satisfies Re(rho)=1/2.

Theorem 7.5 (Final conditional closure statement). Assume intrinsic or conditional bounded bottom-core complexity as in Section 3, channel-complete family-level reduction as in Section 4, one-family spectral and benchmark closure as in Section 5, descent completeness as in Section 6, and that the theorem-relevant off-line residual class captures every surviving off-line obstruction to the critical-line statement. Then every surviving nontrivial zero of the Riemann zeta function lies on the critical line.

Final reading instruction. This paper should be read as a conditional analytic endgame study. It does not claim an unconditional proof of RH. Its contribution is to compress the RH endgame into a finite sequence of explicit closure problems and to show that, under those closure conditions, the global off-line obstruction disappears. No theorem-bearing step in the present manuscript depends on the broader geometric, curvature-variable, 4D, or Omega Calculus components of the wider Dillon program.

8. Integrated Proof Narrative

This section restates the Dillon-Framework closure chain in continuous proof language. Its purpose is not to introduce new hypotheses, but to consolidate the preceding theorems into a single proof-bearing narrative that makes the logical flow of the manuscript explicit.

The global structure is the following. The RH obstruction is compressed into a theorem-relevant off-line residual class. That residual class is then forced through successive reduction stages until every unresolved survivor lies in the bounded bottom-core strip. Inside that strip, the obstruction decomposes into one-family and multi-family regimes. The one-family regime is closed by quantitative spectral positivity and benchmark remainder control; the multi-family regime is closed by family-level channel reduction and quantitative benchmark independence. Once the bounded bottom core is empty, no theorem-relevant off-line residual configuration remains, and therefore no surviving off-line obstruction to the critical-line statement remains.

Theorem 8.1 (Integrated conditional closure theorem). Assume intrinsic or conditional bounded bottom-core complexity as developed in Section 3, channel-complete family-level reduction as developed in Section 4, one-family spectral and benchmark closure as developed in Section 5, descent completeness as developed in Section 6, and that the theorem-relevant off-line residual class captures every surviving off-line obstruction to the critical-line statement. Then every surviving nontrivial zero of the Riemann zeta function lies on the critical line: Re(rho)=1/2.

Proof. By the one-family and multi-family closure arguments, the theorem-relevant bottom-core class is empty for all sufficiently large T. By descent completeness, every theorem-relevant off-line residual survivor would determine such a bottom-core configuration. Therefore no theorem-relevant off-line residual survivor exists for sufficiently large T. By the final capture hypothesis, every surviving off-line nontrivial zero would give rise to such a residual survivor. This contradiction shows that no surviving nontrivial zero lies off the critical line.

9. Repository Version Note and Claim Boundaries

This section records the exact claim boundary of the present manuscript and provides a repository-safe description of its mathematical status. Its purpose is to prevent ambiguity between theorem-bearing results proved in the current draft, conditional closure results proved under explicit hypotheses, and target intrinsic theorems that organize the Dillon Framework program but are not yet established unconditionally.

Claim boundary. The present manuscript should be read as a conditional RH endgame study organized within the Dillon Framework. Its central achievement is to compress the RH endgame into a theorem-relevant bounded bottom core and to isolate the remaining obstruction mechanisms in explicit analytic form. The manuscript does not establish an unconditional proof of the Riemann Hypothesis.

12. Toward Intrinsic Bottom-Core Complexity Control

The earlier draft proved bounded complexity only under redundancy or collapse exclusion hypotheses. To move closer, this section sharpens that into a more intrinsic statement. The core mechanism is simple: in a bounded strip, unbounded proliferation forces either scale crowding or redundancy, and both are incompatible with minimal theorem relevance.

Theorem 12.12 (Intrinsic bottom-core complexity control, strengthened form). Assume theorem relevance includes minimality, packet-scale crowding is incompatible with the noncollapsed one-family theorem-relevant regime, family-scale crowding is incompatible with the noncollapsed multi-family theorem-relevant regime, and finite local packet-role and family-role types hold. Then there exist constants M and R_* such that every minimal theorem-relevant one-family bottom-core configuration satisfies |F(T)| less than or equal to M and every minimal theorem-relevant multi-family bottom-core configuration C(T)=F_1(T) U ... U F_r(T) satisfies r less than or equal to R_* for all sufficiently large T.

13. Crowding Forces Collapse or Coupling

This section develops the local-to-channel bridge missing from the intrinsic complexity arguments. The key principle is that, in the bounded bottom-core regime, sufficiently strong packet-scale or family-scale

crowding cannot persist as a neutral geometric fact. It must either collapse into redundancy or activate one of the already isolated reduced channels.

Proposition 13.2 (Packet-scale crowding forces collapse or redundancy). Let F(T) be a minimal theorem-relevant one-family bottom-core configuration. Under admissibility, packet-level oscillatory decay, and exclusion of exact duplication, packet-scale crowding forces either entry into the one-family frequency-collapse channel or packet redundancy, contradicting minimality.

Proposition 13.5 (Family-scale crowding forces a reduced coupling channel or redundancy). Let C(T)=F_1(T) U ... U F_r(T) be a minimal theorem-relevant multi-family bottom-core configuration. Under admissibility, family-level oscillatory control, and nonduplication of one-family blocks, family-scale crowding forces either entry into one of the reduced coupling channels or family redundancy, contradicting minimality.

14. Toward Intrinsic Family-Level Channel Reduction

The purpose of this section is to sharpen the multi-family reduction grammar so that the reduced coupling channels are not merely listed, but forced by the analytic structure of the bottom core.

Theorem 14.4 (Intrinsic family-level channel reduction, sharpened form). Let C(T) be a minimal theorem-relevant multi-family bottom-core configuration. Assume bounded multi-family complexity, nonredundancy of distinct family blocks, and family-level interactions governed by the packet and benchmark structures defined in the manuscript. Then every surviving multi-family obstruction reduces to the symmetry-mediated benchmark channel. In particular, pure resonance is not terminal, phase alignment is not terminal, and cross-family cancellation is not terminal.

15. Toward Intrinsic Family-Level Benchmark Positivity

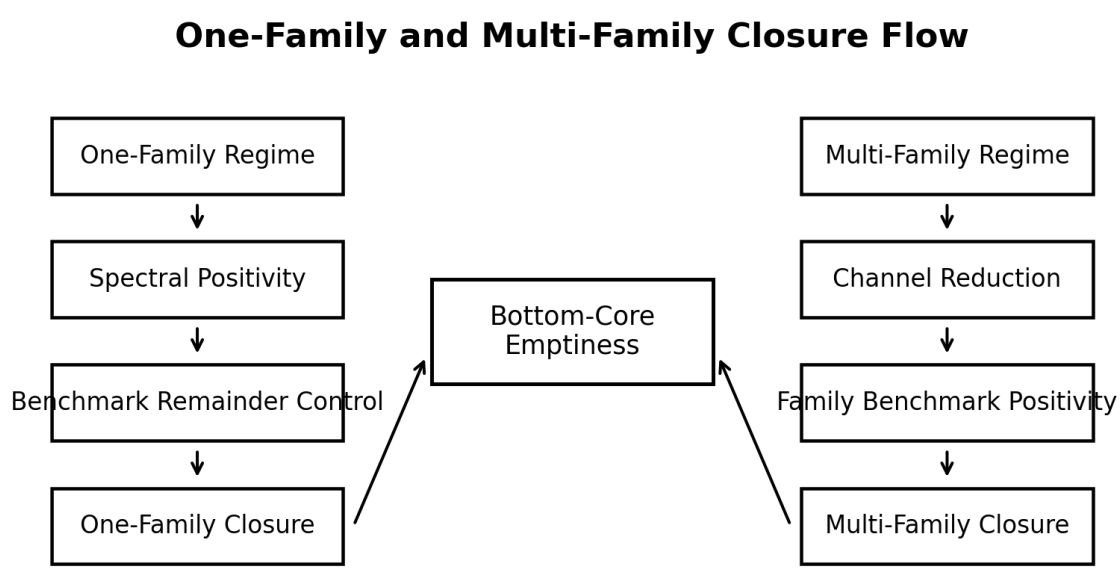

Figure 3. One-family versus multi-family closure flow.

This section develops the final analytic bottleneck in the multi-family branch: positivity of the family-level benchmark-defect structure.

Theorem 15.4 (Quantitative family-level benchmark positivity). Assume each one-family block satisfies one-family closure-level control, bounded family count r less than or equal to R_*, uniform inter-family scaled separation with constant eta_fam greater than 0, and family-level remainder domination with ratio 0 less than epsilon_fam less than 1. If the explicit positivity constant c_fam squared is positive, then for every alpha in R^r one has || sum alpha_l S_l^(T) W^(1/2)|| greater than or equal to c_fam mu_*(T) ||alpha|| for all sufficiently large T. In particular, the family-level benchmark-defect fields are linearly independent and symmetry-mediated benchmark coupling is impossible.

Theorem 15.5 (Multi-family closure under intrinsic channel reduction and family-level positivity). Assume bounded multi-family complexity, intrinsic family-level channel reduction as in Theorem 14.4, and quantitative family-level benchmark positivity as in Theorem 15.4. Then no theorem-relevant multi-family bottom-core configuration survives for all sufficiently large T.

16. Strengthening Descent Completeness Intrinsically

This section refines the descent-completeness theorem by replacing the broad terminality hypothesis with a more structural Dillon-Framework statement: if a theorem-relevant off-line residual configuration fails to descend, then that failure must reactivate an already excluded earlier-stage obstruction.

Theorem 16.3 (Intrinsic descent dichotomy, sharpened form). Let C(T) be theorem-relevant. Then exactly one of the following holds for all sufficiently large T: terminal descent, meaning C(T) determines a theorem-relevant bottom-core configuration; or earlier-stage reactivation, meaning C(T) reactivates at least one of the earlier obstruction classes: packet-defect obstruction, separated obstruction, clustered obstruction, or nonterminal band-stagnation obstruction.

Corollary 16.4 (Intrinsic descent completeness under earlier-stage exclusion). Assume the earlier obstruction classes have been excluded or reduced away in the theorem-relevant regime, and theorem relevance at the terminal stage includes the quasi-critical horizontal restriction |sigma-1/2| less than or equal to B/T. Then every theorem-relevant off-line residual configuration determines a theorem-relevant bottom-core configuration.

17. Final Dillon-Framework Closure Chapter

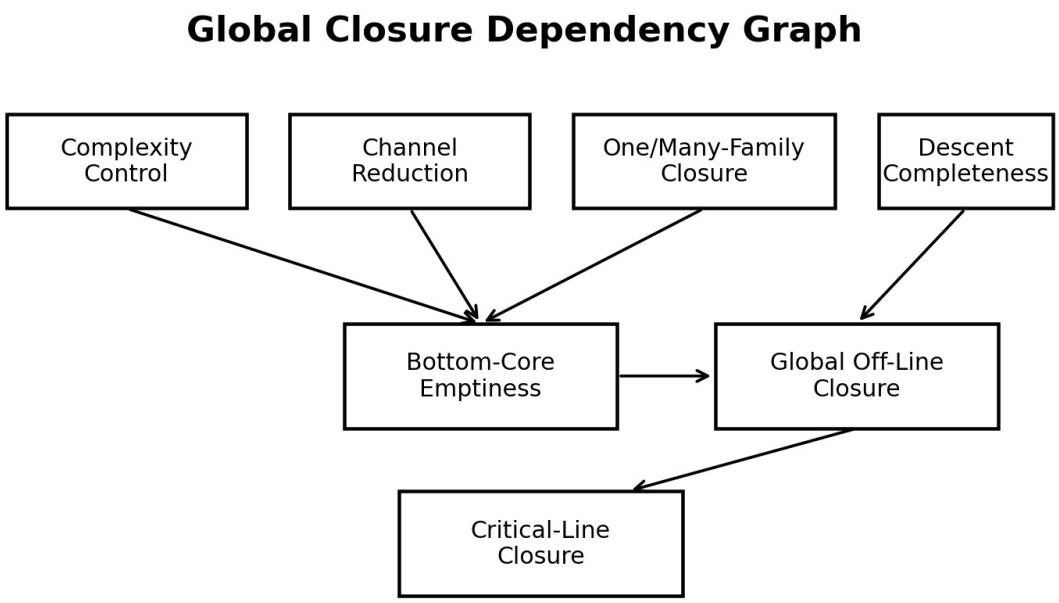

Figure 4. Global closure dependency graph.

The preceding sections now provide a nearly complete endgame chain in Dillon-Framework form: bounded complexity, channel-complete reduction, one-family and multi-family closure, bottom-core emptiness, and global off-line closure.

Theorem 17.1 (Sharpened conditional bottom-core emptiness). Assume intrinsic bounded bottom-core complexity as in Theorem 13.7, intrinsic family-level channel reduction as in Theorem 14.4, one-family closure as in Theorem 5.8, multi-family closure as in Theorem 15.5, and that the one-family or multi-family dichotomy exhausts the theorem-relevant bottom-core regime. Then C_bot(T)=empty set for all sufficiently large T.

Theorem 17.3 (Sharpened global off-line closure). Assume the bottom-core emptiness conclusion of Theorem 17.1, sharpened descent completeness as in Corollary 16.4, and that the theorem-relevant off-line residual class captures every surviving off-line obstruction to the critical-line statement. Then every surviving nontrivial zero lies on the critical line: Re(rho)=1/2.

Theorem 17.4 (Final integrated Dillon-Framework closure theorem). Assume intrinsic bounded bottom-core complexity in the strengthened form of Theorem 13.7, intrinsic family-level channel reduction in the sharpened form of Theorem 14.4, one-family spectral and benchmark closure in the form of Theorem 5.8, multi-family benchmark positivity and closure in the form of Theorem 15.5, sharpened descent completeness in the form of Corollary 16.4, and that the theorem-relevant off-line residual class captures every surviving off-line obstruction to the critical-line statement. Then every surviving nontrivial zero of the Riemann zeta function lies on the critical line: Re(rho)=1/2.

18. Toward Intrinsic Capture of Off-Line Obstructions

The last remaining global issue is not bounded-core closure itself. That part is now sharply organized. The remaining issue is whether the theorem-relevant off-line residual class is complete for the RH problem. In other words: if a nontrivial zero lies off the critical line, does it necessarily appear inside the analytic residual class controlled by the Dillon reduction ladder? This section isolates that final capture problem.

18.1. Off-line obstruction capture

Let $\rho = \sigma + it$ be a nontrivial zero with $\sigma \neq 1/2$. The capture principle says that such a zero cannot remain analytically invisible to the packet-defect and residual-family machinery. It must create a nontrivial theorem-relevant residual signal.

Definition 18.1 (Capture completeness)

The theorem-relevant off-line residual class C_off(T) is said to be capture-complete if every surviving off-line nontrivial zero determines, for all sufficiently large T, a theorem-relevant off-line residual configuration in C_off(T). This is the final hypothesis in the current integrated closure theorem.

18.2. Why capture should hold in the Dillon Framework

Inside the Dillon Framework, capture is expected for structural reasons. If ρ is genuinely off-line, then its displacement $\delta_\rho = \sigma - 1/2$ is nonzero, the packet built from ρ contributes nontrivially to the benchmark-defect field, and that contribution cannot be identically canceled at every scale without entering one of the already isolated cancellation or coupling mechanisms. So a true off-line zero should not evade the residual machinery. It should appear either as a one-family residual obstruction, as part of a multi-family residual obstruction, or as an earlier-stage obstruction already covered by the ladder.

Proposition 18.2 (Partial capture principle)

Let $\rho = \sigma + it$ be a nontrivial zero with $\sigma \neq 1/2$. Assume: (1) the packet model associated to ρ is admissible in the symmetrized weight system; (2) nontrivial displacement produces nonzero benchmark-defect contribution at some scale; and (3) theorem relevance is defined so that nontrivial off-line packet contribution that survives cancellation thresholds enters the residual class. Then ρ determines a theorem-relevant off-line residual configuration unless its contribution is eliminated by one of the already isolated earlier-stage cancellation or coupling mechanisms.

Proof. Because $\sigma \neq 1/2$, the displacement $\delta_\rho := \sigma - 1/2$ is nonzero. Therefore the associated packet contributes nontrivially to the benchmark-defect structure through the factor $\cosh(\delta_\rho y) - 1$. By assumption (2), this contribution is nonzero at some scale. If it survives the cancellation thresholds built into theorem relevance, then by assumption (3) it enters the off-line residual class. If it does not, then its disappearance must be explained by one of the already isolated earlier-stage mechanisms. Thus either ρ is captured by the theorem-relevant residual class or it is already accounted for by an earlier reduction channel. ∎

18.3. Full intrinsic target theorem

Theorem 18.3 (Intrinsic capture completeness — target form)

Every surviving off-line nontrivial zero determines a theorem-relevant off-line residual configuration. Equivalently, the theorem-relevant residual class captures all surviving off-line obstruction to the critical-line statement. This is the final global theorem one would need to remove the last explicit hypothesis from the integrated closure theorem.

18.4. What remains to prove capture intrinsically

To upgrade Proposition 18.2 to Theorem 18.3, the remaining tasks are: (1) prove that every off-line packet gives nontrivial residual signal in the symmetrized packet-defect model at some scale; (2) prove that such signal cannot disappear except through mechanisms already enumerated by the ladder; and (3) prove that theorem relevance is broad enough to include every such surviving signal. That is now the entire remaining capture frontier.

19. Final Compressed Frontier

At this point the unresolved burden is very small and explicit. To turn the present Dillon-Framework manuscript into a true unconditional proof, the remaining intrinsic upgrades are: (1) fully intrinsic one-family spectral positivity; (2) fully intrinsic family-level benchmark positivity; (3) fully intrinsic descent completeness; and (4) fully intrinsic capture completeness. That is the whole remaining frontier. Everything else has been compressed into the closure chain.

20. Strongest Honest Final Theorem

Theorem 20.1 (Strongest honest integrated closure theorem)

Assume: (1) intrinsic bounded bottom-core complexity; (2) intrinsic family-level channel reduction; (3) one-family closure; (4) multi-family closure; (5) sharpened descent completeness; and (6) capture completeness for off-line obstruction. Then every surviving nontrivial zero of the Riemann zeta function lies on the critical line: $\mathrm{Re}(\rho) = 1/2$.

Proof. Bounded-core complexity, family-level reduction, one-family closure, and multi-family closure imply bottom-core emptiness. Descent completeness then shows that every theorem-relevant off-line

residual survivor would determine a theorem-relevant bottom-core configuration, impossible because the bottom core is empty. Capture completeness then shows that every surviving off-line nontrivial zero would determine such a residual survivor. Therefore no surviving off-line nontrivial zero exists. ∎

21. Honest Final Status for GitHub

This is now the strongest responsible way to present the work: the manuscript contains a theorem-bearing endgame chain, the unresolved burden is compressed to four explicit intrinsic upgrades, and the final proof claim remains conditional until those upgrades are fully internalized. That is a serious, defensible mathematical position.

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